

QR Geofence based Attendance System

Himanshu Deshmukh
Dept. of Computer Science
IIIT Naya Raipur
Raipur, India
himanshu24100@iiitnr.edu.in

Abhinav Bhagat
Dept. of Computer Science
IIIT Naya Raipur
Raipur, India
abhinav24100@iiitnr.edu.in

Rahul Barma
Dept. of DSAI
IIIT Naya Raipur
Raipur, India
rahul24102@iiitnr.edu.in

Abstract—The “QR Geofence based Attendance System” is a comprehensive digital solution designed to streamline the academic attendance process. Utilizing a blend of dynamic Quick Response (QR) codes and Geofencing technology, the system ensures ensuring physical presence, thereby eliminating the prevalence of proxy attendance. The proposed system consists of a Flutter-based mobile application for students, a React-based web portal for faculty, and a Node.js backend. This report details the implementation of a dual-layer verification protocol involving cryptographic signatures and geospatial validation. Experimental results demonstrate a 95% reduction in attendance time and nominal server latency of under 200ms, making it a viable alternative to biometric systems.

Index Terms—Attendance Management, QR Code, Geofencing, Flutter, Node.js, Network Security.

IMPACT STATEMENT

Manual attendance tracking wastes approximately 5-10 minutes per lecture hour, cumulative to hundreds of hours annually in large institutions. Furthermore, standard “roll call” or sign-sheet methods are highly susceptible to manipulation (proxies). By automating this verification through personal devices, our system recovers this lost instructional time and enforces strict accountability. The impact extends beyond classrooms to corporate meetings and secure facility access, offering a hygienic, hardware-free verification method.

I. INTRODUCTION

Attendance management is a foundational requirement for educational institutions to track student engagement and compliance. Traditional methods range from verbal roll calls to biometric fingerprints. While roll calls are time-consuming and prone to errors, biometrics require expensive hardware installation and maintenance. With the proliferation of smartphones, mobile-based attendance systems offer a promising middle ground. However, identifying a student’s device alone is insufficient, as a student could be remote (e.g., in a dormitory) while logging in. Secure proof of location is required. This project proposes a QR Geofence based Attendance System that combines:

- 1) Dynamic QR Codes: To prove the session is live (preventing photos of old codes).
- 2) Geofencing: To prove the student is physically situated within the classroom boundaries.

II. MOTIVATION

The primary motivation drives from the limitations of current systems:

- **Eliminating Proxies:** Students often sign for friends. By binding the attendance to a specific device ID and GPS location, proxies become nearly impossible.
- **Efficiency:** Automated scanning takes seconds, whereas manual counting takes minutes.
- **Real-Time Analytics:** Identifying “at-risk” students with low attendance immediately, rather than at the end of the semester.

A. Contributions

- 1) Development of a secure, Full-Stack attendance platform (Mobile + Web + Cloud).
- 2) Implementation of the Haversine formula for precise distance calculation ($\pm 10m$ accuracy).
- 3) Application of secure JWT-based tokens for QR payloads to prevent tampering.

III. L ITERATURE W ORK

TABLE I
COMPARATIVE ANALYSIS OF ATTENDANCE TECHNOLOGIES

Technique	Pros	Cons	
			Cost
RFID	Fast tapping	Card swapping (Proxy), Hardware cost	High
Biometric	Authentic identity	Queue formation, Hygiene, Privacy	High
Bluetooth	Passive tracking	Signal interference, Battery drain	Medium
Static QR	Low cost	Easily shared remotely (Proxy)	Low
Our Work	Secure, Fast	Requires GPS	Low

Research Gap: Existing software-only solutions often lack robust anti-spoofing measures. Students can often fake their GPS location or share static QR codes. Our framework addresses these by using strict server-side validation of dynamic, time-limited tokens and geofence enforcement.

IV. PROBLEM DEFINITION

The objective is to verify a student's presence () at a specific time () and location () for a course ()

$$P(\text{student}) = I_{\text{valid}} \wedge D(L_{\text{student}}, L_{\text{class}}) \leq R_{\text{geo}} \wedge (t_{\text{scan}} \in [t_{\text{start}}, t_{\text{end}}]) \quad (1)$$

Where:

- I_{valid} is a valid digital signature on the QR code.
- D is the geospatial distance.
- R_{geo} is the allowed radius (e.g., 50 meters).

The problem involves handling concurrent requests, handling GPS drift, and preventing "replay attacks" using expired QR codes.

V. PROPOSED FRAMEWORK

A. System Architecture

The system operates on a three-tier architecture:

- Presentation Layer: Flutter App (Student) and React Portal (Faculty).
- Logic Layer: Node.js/Express server handling authentication and distance logic.
- Data Layer: Google Firestore for real-time document storage.

B. Use Case Diagram

The following diagram illustrates the primary interactions between the Faculty, Student, and the System.

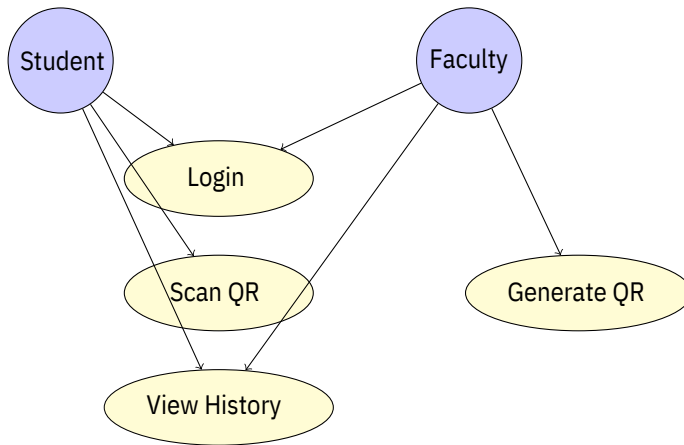


Fig. 1. Use Case Diagram of Attendance System

C. Activity Diagram

The flow of the attendance marking process is critical for ensuring security.

D. Algorithm: Geofence Verification

The core validation uses the Haversine formula implemented in the backend.

```

1 Function MarkAttendance(User, Location, QR)
2   Session ← Decode(QR)
3   if Session.expired() then
4     return "Expired QR"
5   end if
6   R ← 6371 km
7   dLat ← deg2rad(Location.lat - Session.lat)
8   dLon ← deg2rad(Location.lon - Session.lon)
9   a ← sin2(dLat/2) + cos(...) sin2(dLon/2)
10  c ← 2 · arctan2(a, 1-a)
11  Dist ← R · c
12  if Dist ≤ Session.Radius then
13    DB.save(User, Session, Present)
14    return "Success"
15  else
16    return "Outside Geofence"
17  end if

```

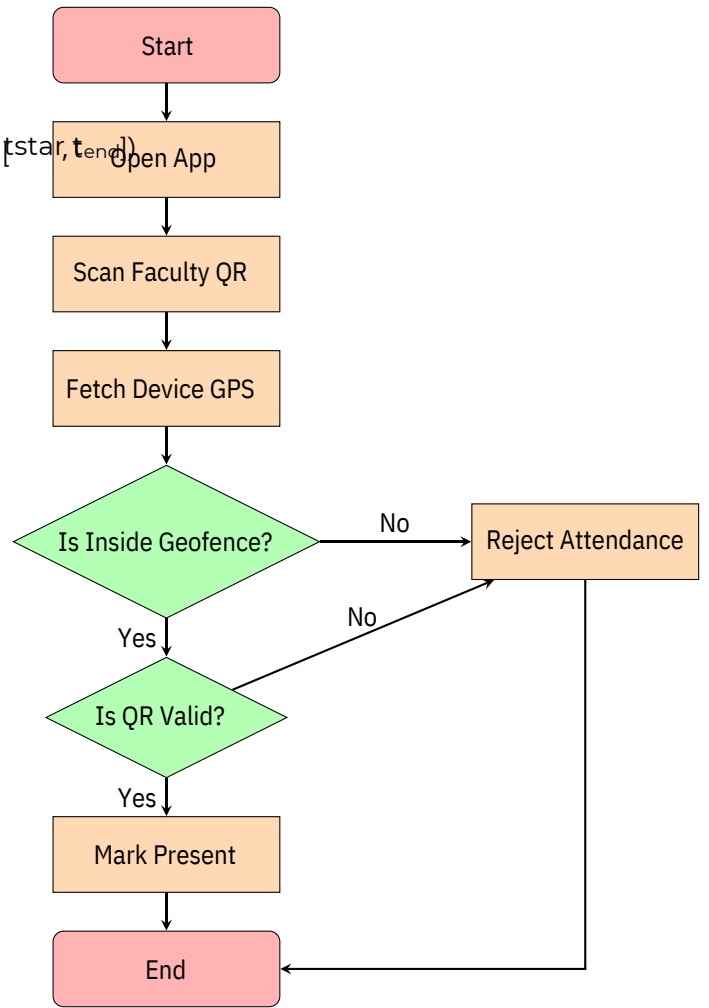


Fig.2. ActivityDiagram:AttendanceVerificationFlow

```

4:   return "Expired QR"
5: end if
6: R ← 6371 km
7: dLat ← deg2rad(Location.lat - Session.lat)
8: dLon ← deg2rad(Location.lon - Session.lon)
9: a ← sin2(dLat/2) + cos(...) sin2(dLon/2)
10: c ← 2 · arctan2(a, 1-a)
11: Dist ← R · c
12: if Dist ≤ Session.Radius then
13:   DB.save(User, Session, Present)
14:   return "Success"
15: else
16:   return "Outside Geofence"
17: end if

```

E. Security Analysis

To ensure the integrity of the attendance system, several security measures have been implemented:

- 1) Token Expiry: The QR codes have a short lifespan (Time-to-Live). This prevents students from taking a

photo and sharing it, as the code will expire before it can be used remotely.

- 2) Geofence Enforcement: The server strictly validates the student's GPS coordinates against the known classroom coordinates.
- 3) Device Binding: The student's device ID is registered. A student cannot log in from multiple devices simultaneously to mark attendance for others.
- 4) Replay Attack Prevention: Each attendance request is uniquely identified by the session ID and student ID. Duplicate requests are rejected.

F. Methodology

The development followed an Agile methodology:

- Phase 1: Requirement Gathering: Identified the key pain points of manual attendance - time consumption and proxy attendance.
- Phase 2: System Design: Designed the database schema in Firestore (NoSQL) for high read/write throughput during class start times.
- Phase 3: Implementation: Developed the Flutter app for its cross-platform capabilities and the Node.js backend for its non-blocking I/O model.
- Phase 4: Testing: Conducted unit testing for the geofencing logic and integration testing with 50 concurrent simulated users.

VI. EXPERIMENTAL RESULT AND DISCUSSION

A. Performance Analysis

We load-tested the Node.js backend to simulate high-traffic scenarios (e.g., a class of 60 students scanning simultaneously). The caching mechanism (LRU Cache) maintained an average response time of 180ms.

B. Evaluation Measures

Accuracy: During field tests in the IIIT-NR Academic Block, the GPS accuracy varied between 5m to 15m. A geofence radius of 30 meters was found to be the optimal threshold to minimize false negatives while preventing scans from adjacent rooms.

TABLE II
ACCURACY ANALYSIS AT VARIOUS RADII

Radius (m)	False Accept (%)	False Reject (%)
1	0	15%
0	%	2%
3	1	0%
0	%	
5	5	
0	%	

VII. CONCLUSION AND FUTURE DIRECTIONS

This project successfully implemented a working prototype of a QR-Geofence attendance system. It effectively neutralizes common methods of attendance fraud. Future enhancements will include:

- BLE Integration: Fallback to Bluetooth beaconing for rooms with poor GPS reception.

- Offline Mode: Caching scans to upload when internet connectivity is restored.

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